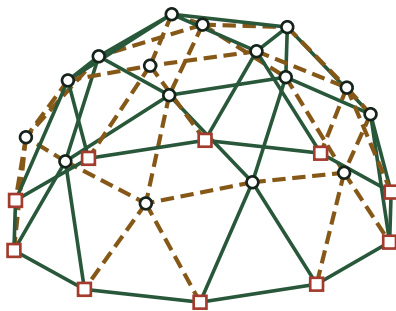


PREPARED FOR JANTZ / BLACK BELT BUILDING

12 FT 2V WOOD DOME

Audited geometry + field planning reference

G-000 CANONICAL FRAME AXONOMETRIC - NOT TO SCALE



MEMBER LEGEND --- SHORT — LONG

REVIEW LAYER	STATUS	BOUNDARY
Canonical shell	VERIFIED	26 nodes / 65 member axes / 40 faces
Gross wood faces	VERIFIED	30 P1 + 10 P2 node-center templates
Joinery	SPATIAL CHECK ONLY	Digital clearance; strength and retention open
Entrance + platform	CONCEPT ONLY	Replacement load path and foundation open

REFERENCE STUDY - NOT FOR FABRICATION, STRUCTURAL, PERMIT, OR OCCUPANCY USE

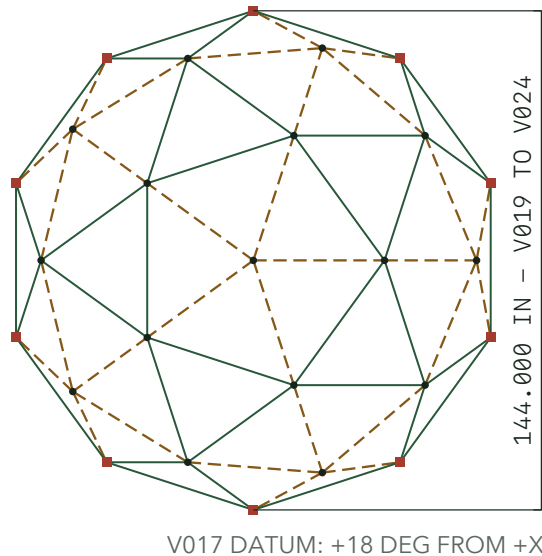
Print-optimized and text-searchable. Accessible interactive companion: monroe-domes.vercel.app. This PDF is not represented as PDF/UA tagged.

G-101 / CANONICAL MODEL

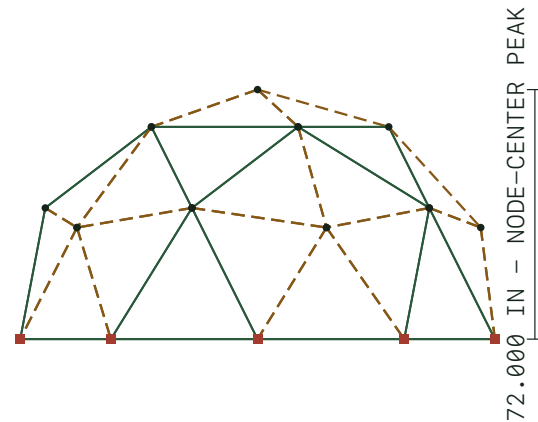
Plan and elevation

All linework terminates at mathematical node centers. Dimensions describe the computational axes, not timber outside faces or interior clearances.

A-101 PLAN - NODE CENTERLINES - NTS



A-201 FRONT ELEVATION - NTS



Base node plane: Y = 0.000 in

PARAMETER	AUDITED VALUE	MEANING
Sphere radius	72.000 in	Center to mathematical node sphere
Base diameter	144.000 in	V019 to V024; opposite base nodes
Peak height	72.000 in	Base node plane to apex node
Base	Regular decagon	10 coplanar nodes / 10 boundary chords

Checks: $V - E + F = 26 - 65 + 40 = 1$. Port incidence: $2E = 130 = 10(4) + 6(5) + 10(6)$. +Y is vertical; the base lies at Y = 0. V017 is +18 DEG from +X at (68.476069, 0.000000, 22.249224) in.

FIGURE TEXT EQUIVALENT: The plan is a regular decagonal hemispherical triangulation. The elevation rises from a planar ten-node base to one apex at 72 inches.

G-201 / PARTS BEFORE ASSEMBLY

Every timber axis and node kit

Piece IDs match the interactive model. Every timber length below is node-center geometry only - not a blank, shoulder, tip-to-tip, or cut length.

FAMILY	QTY	NODE-CENTER AXIS	INCIDENCE
Short S	30	39.350380 in	10 H4-H5 + 20 H5-H6
Long L	35	44.498447 in	10 H4-H4 + 10 H4-H6 + 15 H6-H6
Node kits	26	10 H4 / 6 H5 / 10 H6	Topology families; not machining templates

SHORT PIECES	LONG PIECES	NODE KITS
S-01 V001-V002	L-01 V002-V003	V001 H5 ABOVE
S-02 V001-V003	L-02 V002-V006	V002 H6 ABOVE
S-03 V001-V004	L-03 V002-V007	V003 H6 ABOVE
S-04 V001-V005	L-04 V002-V011	V004 H6 ABOVE
S-05 V001-V006	L-05 V003-V004	V005 H6 ABOVE
S-06 V002-V012	L-06 V003-V007	V006 H6 ABOVE
S-07 V003-V013	L-07 V003-V008	V007 H6 ABOVE
S-08 V004-V014	L-08 V004-V005	V008 H6 ABOVE
S-09 V005-V015	L-09 V004-V008	V009 H6 ABOVE
S-10 V006-V016	L-10 V004-V009	V010 H6 ABOVE
S-11 V007-V012	L-11 V005-V006	V011 H6 ABOVE
S-12 V007-V013	L-12 V005-V009	V012 H5 ABOVE
S-13 V008-V013	L-13 V005-V010	V013 H5 ABOVE
S-14 V008-V014	L-14 V006-V010	V014 H5 ABOVE
S-15 V009-V014	L-15 V006-V011	V015 H5 ABOVE
S-16 V009-V015	L-16 V007-V017	V016 H5 ABOVE
S-17 V010-V015	L-17 V007-V018	V017 H4 BASE
S-18 V010-V016	L-18 V008-V019	V018 H4 BASE
S-19 V011-V012	L-19 V008-V020	V019 H4 BASE
S-20 V011-V016	L-20 V009-V021	V020 H4 BASE
S-21 V012-V017	L-21 V009-V022	V021 H4 BASE
S-22 V012-V026	L-22 V010-V023	V022 H4 BASE
S-23 V013-V018	L-23 V010-V024	V023 H4 BASE
S-24 V013-V019	L-24 V011-V025	V024 H4 BASE
S-25 V014-V020	L-25 V011-V026	V025 H4 BASE
S-26 V014-V021	L-26 V017-V018	V026 H4 BASE
S-27 V015-V022	L-27 V017-V026	
S-28 V015-V023	L-28 V018-V019	
S-29 V016-V024	L-29 V019-V020	
S-30 V016-V025	L-30 V020-V021	
	L-31 V021-V022	
	L-32 V022-V023	
	L-33 V023-V024	
	L-34 V024-V025	
	L-35 V025-V026	

MATERIAL BASIS: Douglas fir, nominal 2 x 2 in, modeled dressed section 1.500 x 1.500 in. Confirm actual stock, moisture, grade, defects, and grain before any fabrication release.

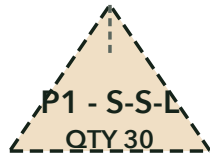
P-101 / GROSS WOOD FACE SCHEDULE

Two exact face families

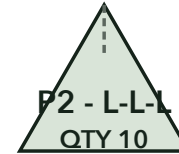
These triangles are derived from the canonical node-center faces. Gross total: 30238.094167 sq in / 209.986765 sq ft. They are not finished panel cuts or purchase quantities.

P-101 GROSS NODE-CENTER FACE FAMILIES - NOT CUT TEMPLATES

SIDES 39.350380 / 39.350380 IN
BASE 44.498447 · ALT 32.456501 IN



SIDES 44.498447 IN EACH
ALT 38.536786 IN



GROSS MODEL FACES ONLY - EDGE BUILD-UP, CUTBACKS, JOINTS, GASKETS, AND DRAINAGE ARE UNSET

TYPE	QTY	GROSS SIDES	ANGLES	AREA EACH
P1 S-S-L	30	39.350380 / 39.350380 / 44.498447 in	55.569012 / 55.569012 / 68.861976 DEG	722.131952 sq in
P2 L-L-L	10	44.498447 in equilateral	60 / 60 / 60 DEG	857.413562 sq in

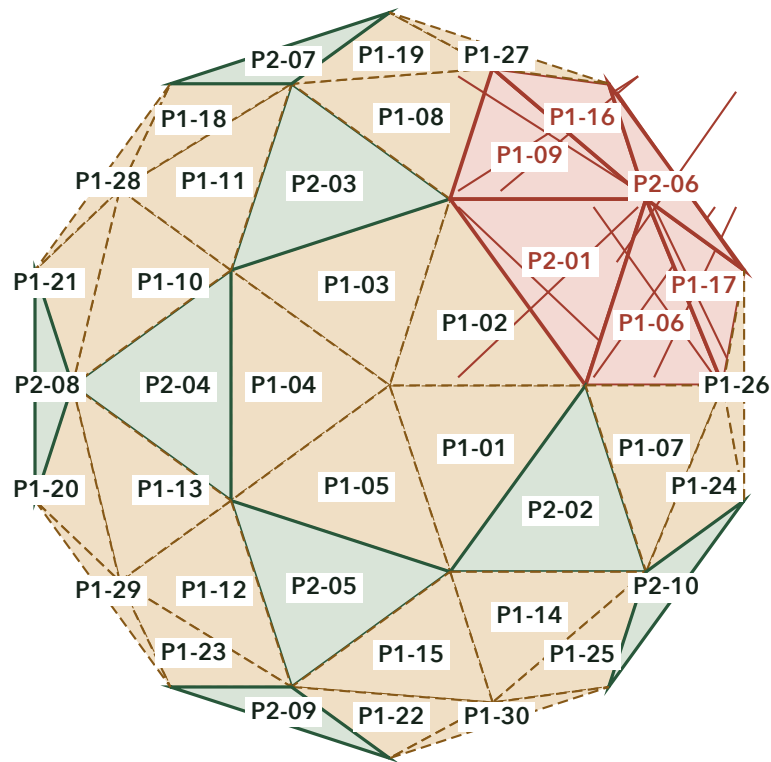
P1 PIECE / FACE / VERTICES	P2 PIECE / FACE / VERTICES
P1-01 F001 V001-V002-V006	P2-01 F006 V002-V003-V007
P1-02 F002 V001-V003-V002	P2-02 F008 V002-V011-V006
P1-03 F003 V001-V004-V003	P2-03 F010 V003-V004-V008
P1-04 F004 V001-V005-V004	P2-04 F013 V004-V005-V009
P1-05 F005 V001-V006-V005	P2-05 F016 V005-V006-V010
P1-06 F007 V002-V007-V012	P2-06 F023 V007-V018-V017
P1-07 F009 V002-V012-V011	P2-07 F026 V008-V020-V019
P1-08 F011 V003-V008-V013	P2-08 F029 V009-V022-V021
P1-09 F012 V003-V013-V007	P2-09 F032 V010-V024-V023
P1-10 F014 V004-V009-V014	P2-10 F035 V011-V026-V025
P1-11 F015 V004-V014-V008	
P1-12 F017 V005-V010-V015	
P1-13 F018 V005-V015-V009	
P1-14 F019 V006-V011-V016	
P1-15 F020 V006-V016-V010	
P1-16 F021 V007-V013-V018	
P1-17 F022 V007-V017-V012	
P1-18 F024 V008-V014-V020	
P1-19 F025 V008-V019-V013	
P1-20 F027 V009-V015-V022	
P1-21 F028 V009-V021-V014	
P1-22 F030 V010-V016-V024	
P1-23 F031 V010-V023-V015	
P1-24 F033 V011-V012-V026	
P1-25 F034 V011-V025-V016	
P1-26 F036 V012-V017-V026	
P1-27 F037 V013-V019-V018	
P1-28 F038 V014-V021-V020	
P1-29 F039 V015-V023-V022	
P1-30 F040 V016-V025-V024	

P-102 / PANEL PLACEMENT

All 40 faces in one map

The top projection is a non-overlapping map of the canonical triangulated disk. Each printed label is the panel piece ID listed on P-101.

P-102 40-FACE PLAN PLACEMENT MAP - NOT TO SCALE



P1 = dashed isosceles

P2 = solid equilateral

X-MARKED = entrance-view omitted

ENTRANCE-VIEW AFFECTED FACES	PIECE CONSEQUENCE	STATUS
F006 / F007 / F012 / F021 / F022 / F023	Six canonical face panels hidden only in entrance view	No replacement enclosure designed

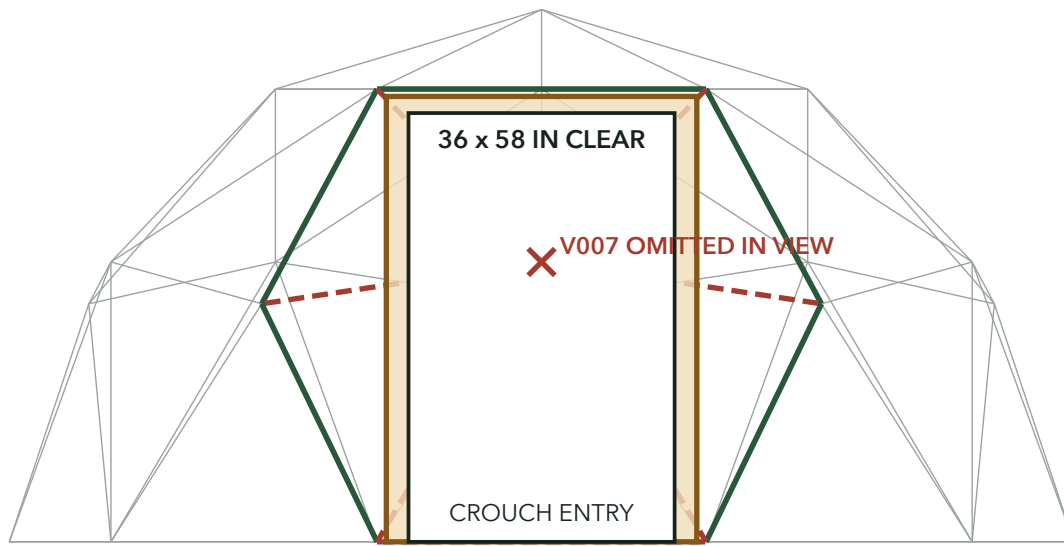
FIGURE TEXT EQUIVALENT: Thirty isosceles P1 faces and ten equilateral P2 faces tile the complete dome. Six marked faces form the optional entrance visualization patch.

E-101 / OPTIONAL ENTRANCE STUDY

Crouch opening aligned to one shell patch

The entrance is a visual opening study. It deliberately does not alter the canonical schedules or exports; affected shell parts are hidden only in this view.

E-101 OPTIONAL ENTRANCE PATCH + CASSETTE - REPLACEMENT LOAD PATH NOT DESIGNED



DASHED = omitted canonical members

HEAVY SOLID = retained patch boundary

ITEM	VALUE	BOUNDARY
Clear opening	36 x 58 in	Crouch entry; not full standing access
Outer cassette	42.00 x 60.25 in	3 in jambs / 2.25 in lintel study
Source patch rise	61.246858 in	Geometric patch bound only
Hidden members	L-03 / L-06 / S-11 / S-12 / L-16 / L-17 / L-26	Seven canonical members omitted in view
Retained boundary	L-01 / S-06 / S-07 / S-21 / S-23	Five canonical boundary members remain
Hidden node	V007	One canonical node omitted in view

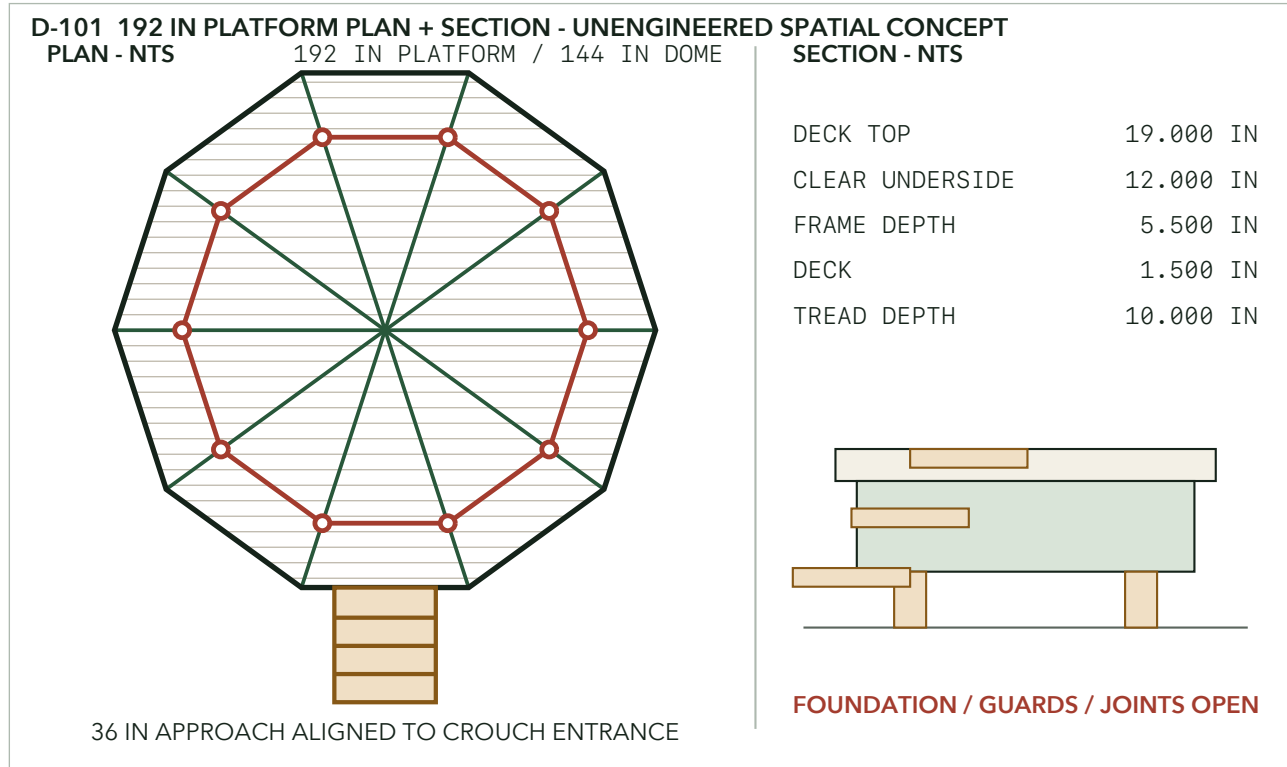
VISUAL OPENING STUDY - REPLACEMENT LOAD PATH NOT DESIGNED. Reinforcement, joints, door, threshold, weather seals, drainage, egress, guards, foundation, and acoustic performance remain unevaluated.

FIGURE TEXT EQUIVALENT: Dashed red members and node V007 are omitted from the entrance view. A 36 by 58 inch clear cassette is centered within the selected six-face patch.

D-101 / PLATFORM + APPROACH

Dome above a larger all-wood platform concept

The platform is spatially aligned with the 144 inch dome and 36 inch entrance. It is not a joist, footing, guard, or connection design.



PARAMETER	VALUE	MODEL MEANING
Platform diameter	192.000 in	Regular decagon, corner to corner
Dome base diameter	144.000 in	Node center, corner to corner
Apron	24.000 in vertex / 22.825356 in flat	Not 24 in everywhere
Outer side	59.331263 in	Ten equal rim segments
Deck + frame	35 boards / 10 radial / 10 rim	Counts only; sizes unengineered
Supports	10 + one center hub	Concept axes only
Vertical	19.000 in deck top / 12.000 in clear underside	Three-rise approach; 10 in treads

ALL-WOOD CONNECTION GOAL ONLY. The H4 clearance body reaches 1.500 in below the base-node datum, so a recess or raised bearing datum is unresolved. Foundation, load path, isolation, weathering, guards, joints, and acoustics are not modeled.

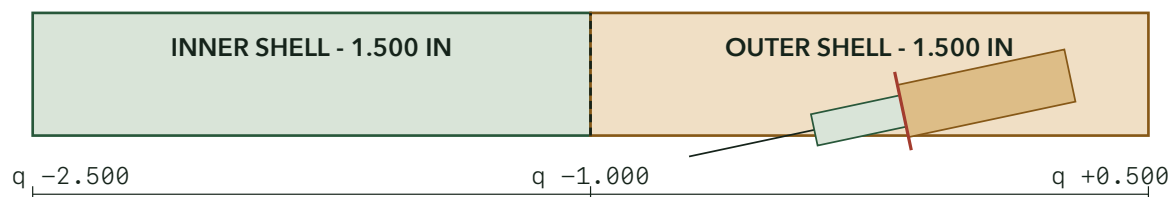
FIGURE TEXT EQUIVALENT: A 192 inch decagonal deck surrounds the 144 inch dome, with ten radial frame lines, ten supports at dome base-node axes, and a centered three-step approach.

J-301 / PORT-NORMAL CLEARANCE

Spatial fit is closed; structural design is open

Each shoulder face is normal to its member axis at one fixed setback. The two-shell body and pocket volumes are explicit digital envelopes, not selected or capacity-checked joinery.

CURRENT JOINT STUDY

**SPATIAL CLEARANCE VERIFIED - STRUCTURAL REVIEW
OPEN****J-301 PORT-NORMAL HUB ENVELOPE - SPATIAL CLEARANCE ONLY****SHOULDER FACE NORMAL TO MEMBER AXIS****TENON + OVERSIZED POCKET ENVELOPE****SCHEMATIC DATUM RELATIONSHIP - NOT A JOINT DETAIL**

INPUT	VALUE	AUDIT MEANING
Shoulder	Axis-normal at S = 4.000 in	Fixed port datum
Convex footprints	H4 9.693632 / H5 10.110635 / H6 10.270510 in	Maximum planar diameters
Tenon study	1.250 x 0.750 x 0.500 in	Integral volume only
Pocket envelope	1.300 x 0.780 x 0.530 in	Collision envelope; not fit tolerance
Radial slab	q = -2.500 to +0.500 in	Two 1.500 in shells split at q = -1.000
Member roll	width-radial · fixed	Fixed study condition

CLEARANCE CHECK	RESULT	MINIMUM
Sampled pocket pairs	0 / 24,570 collide	1.396637 in
Sampled member pairs	0 / 24,570 collide	1.537727 in
Continuous pocket bound	0 / 270 collide	1.380331 in
Continuous member bound	0 / 270 collide	1.408257 in
Pocket split	Every pocket crosses q = -1	0.347980 in each side
Shoulder to other faces	Every modeled shoulder fits	1.011118 in

SUPERSEDED HISTORY: The earlier untrimmed 1.5 x 1.25 x 0.5 in tenon-box layout intersected at every tested H4, H5, and H6 pair. It was rejected and is not a fabrication option.

No mortise fit, retention, ligament, grain, adhesive, tolerance, moisture, strength, load, assembly, or durability approval is issued.

R-001 / RELEASE GATE + AUDIT TRACE

What Jantz can trust - and what remains open

Rev 07 consolidates the complete digital reference for Black Belt Building without converting planning geometry into construction authorization.

VERIFIED / RECORDED	OPEN / WITHHELD
Class-I 2V topology, node coordinates, 65 axes, two chord classes, 26 node valences, and 40 face incidences.	Structural loads, member capacity, buckling, connection capacity, anchorage, foundation, code, permit, and occupancy.
Gross P1/P2 triangle geometry and 209.986765 sq ft total mathematical face area.	Finished panel cuts, thickness, attachment, seals, drainage, fire, weathering, moisture movement, and acoustics.
Entrance affected-set adjacency and 36 x 58 in visual opening envelope.	Replacement shell load path, reinforcement, joints, door, threshold, egress, guards, and weather enclosure.
192 in platform spatial geometry, apron relationships, frame/support counts, and approach alignment.	Joist/support sizing, bearing, footings, guards, all-wood joint dimensions, isolation, durability, and acoustic tuning.

Model assumptions

- All dimensions are generated from a 72 in spherical radius.
- Member lengths are node-center to node-center chords, not finished stock cuts.
- Rendered hub prisms are schematic topology markers; the full-length timber solids overlap them by construction.
- The current study uses exact port-normal faces at a 4 in axial shoulder setback.
- The two-shell joint study clears the modeled member, tenon, pocket, key-relief, and center-bore envelopes, but is not structurally approved or build-ready.
- The geometric cap is complete; loads, connections, foundations, openings, and code compliance are outside this model.
- The 2 × 2 section is modeled at a dressed 1.5 × 1.5 in and must be checked against the actual stock.

Next gate

Confirm actual stock, species and grade, service moisture and exposure, site loads and occupancy, foundation and anchorage, hub material and layup, retention, assembly sequence, weather enclosure, acoustics, and tooling. Then complete exact solids, ligament and grain checks, engineered capacity calculations, a full-scale prototype/test program, and an explicit fabrication release.

References + trace

Canonical sources: lib/geodesic.ts, lib/joinery.ts, and lib/spec.ts. Serialized by scripts/export_pdf_data.mjs; rendered by scripts/generate_audit_pdf.py. Model-data SHA-256: 4a35b3983a4caca29e24e5ad7462b75609b30482a6bf6f1a7ac64b513348a8eb.

Live accessible model: monroe-domes.vercel.app. Timber reference: TFEC standards. Project-specific engineering remains required.

REVISION	DATE	RECORD
06	2026-08-27	Core centerline and joint-clearance audit
07	2026-08-29	Black Belt Building field reference: parts, panels, entrance, platform, and current release gate

NO APPROVAL OR SIGNATURE BLOCK. REFERENCE STUDY ONLY.